

Pooja Kiran Bharadwaj

Security Engineer | AI Security | Application Security | Cloud & IAM Security

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poojakira.github.io/Pooja_Kiran_Portfolio_Website

SUMMARY

Security engineer who builds and validates controls for AI agents, cloud identities, and model supply chains. Hands-on work spans MCP/JSON-RPC agent security, prompt-injection detection, AWS IAM analysis, and adversarial ML, backed by reproducible tests, CI, and SARIF telemetry in Python.

EXPERIENCE

Independent AI Security Researcher & Engineer

Aug 2024 – Present

Self-Directed Security Research

Tempe, AZ

- Designed and validated security controls across six areas: agent runtime security, cloud IAM, LLM red teaming, model supply-chain security, training-data integrity, and adversarial ML.
- Engineered an MCP/JSON-RPC security gateway with 55 prompt-injection patterns and an AWS IAM analyzer with 25 deterministic rules covering `iam:PassRole`, privilege escalation, trust-policy weaknesses, and audit tampering.
- Backed every claim with reproducible evidence: 659 gateway tests (82% coverage), 235 IAM analyzer tests, and 173 LLM security tests, wired to CI, SARIF output, and audit telemetry.

Business & Compliance Lead, AEROSEC

Aug 2025 – Dec 2025

Honeywell Aerospace Technologies × Arizona State University

Tempe, AZ

- Led compliance and third-party security strategy for AEROSEC, a proposed protection layer addressing operational and security risk across airline Passenger Service System integrations.
- Developed a \$120K first-year commercialization scenario and a five-year financial model, then presented the security and business case to ASU and Honeywell stakeholders.

Graduate Teaching Assistant, IT Grader

Jan 2025 – Oct 2025

Ira A. Fulton Schools of Engineering, Arizona State University

Mesa, AZ

- Evaluated roughly 85 undergraduate web-development submissions per semester and 52 graduate assignments on IT security policy, secure coding, and risk analysis, delivering targeted technical feedback with faculty.

SECURITY ENGINEERING PROJECTS

MCP Agent Security Gateway | *Python, FastAPI, JSON-RPC* · poojakira/mcp-agent-security-gateway

- Built an inline security control that inspects agent MCP tool calls before execution, applying 55 prompt-injection patterns, capability and trust checks, PII/exfiltration signals, and process/egress policy at the agent-to-tool boundary.
- Verified behavior with 659 passing tests and 82% statement coverage; added hash-chained audit logging, rate limiting, 9 Elastic rules, and 21 SIEM tests. Enforcement applies only to routed traffic (research prototype).

AWS Agent Identity Guard | *Python, AWS IAM, SARIF* · poojakira/aws-agent-identity-guard

- Developed a static IAM analyzer for AI-agent roles with 25 deterministic rules for wildcard grants, `iam:PassRole`, `sts:AssumeRole`, privilege escalation, trust-policy risks, and missing permission boundaries.
- Confirmed with 235 passing tests; emitted SARIF 2.1.0 with CI exit codes for GitHub Code Scanning gating. Static analysis only; does not compute effective permissions or enforce at runtime.

HF Model Provenance Scanner | *Python, Pickle/ONNX/Keras, AST* · poojakira/hf-model-provenance-scanner

- Engineered a non-executing model supply-chain scanner using custom pickle-opcode parsing plus AST, taint, and symbolic-string analysis across five artifact formats, mapping findings to MITRE ATT&CK v19.
- Detected 33/33 committed adversarial fixtures with zero actionable findings on the benign set; fails loud on unanalyzable streams. Fixture-scoped result, not a real-world detection rate.

TECHNICAL SKILLS

Security: AI/LLM Security, Application Security, Threat Modeling, AWS IAM, Agent Security, Model Supply-Chain Security, AI Red Teaming

Engineering: Python, FastAPI, REST, JSON-RPC, pytest, GitHub Actions, CI/CD, Docker, Kubernetes, Git

Security Tooling: SARIF 2.1.0, GitHub Code Scanning, CodeQL, Elastic/ECS, SIEM, Prometheus, Audit Logging

EDUCATION

Arizona State University

May 2026

Master of Science in Information Technology (Security), GPA 3.87/4.00

Tempe, AZ

M. S. Ramaiah University of Applied Sciences

Aug 2023

B.Tech in Computer Science & Engineering, CGPA 8.44/10

Bengaluru, India

CERTIFICATIONS & PUBLICATION

AWS Academy: Cloud Security Foundations (Nov 2025), Cloud Architecting (Apr 2025)

Publication: A Personalized E-Learning System Using Reinforcement Learning Through Satellite, IEEE INDICON 2023